Gurpreet Singh

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PROJECTS

TrayMate - AI-driven meal ordering app for care facilities that personalizes dietary compliance

- Engineered secure multi-user workflows and **4 distinct** role-based access levels (**RBAC**) through a production-grade backend using **Spring Boot** and **JWT**.
- Architected and managed a high-integrity 3rd-Normal Form (**3NF**) relational database in **AWS MySQL**, ensuring data consistency across **10+ interconnected tables**.
- Automated deployment and validation through **CI/CD** pipelines, reducing manual testing overhead by **40%** via **GitHub Actions** and automated tests such as **Cypress** to ensure 100% reliability in critical meal-selection flows.

Glowlight - Raspberry Pi Climate Monitoring System

- Built a real-time climate monitoring system that provides instant visual feedback through the integration of a **DHT11 sensor** and an RGB LED using **Python** and **Raspberry Pi**.
- Programmed a dual-mode UI using **GPIO** state logic to manage **10+ hardware states**, allowing users to toggle between Edit and Sense modes seamlessly.
- Mapped **potentiometer** inputs to digital **LCD** limits, keeping **98% accuracy** in hardware-software signaling.

GQueue - Real-Time Flash Sale Engine

- **Built REST APIs and Kafka event pipelines** using Spring Boot 3, separating order requests from inventory processing to speed up API response times.
- **Prevented race conditions and stock overselling** during high traffic by implementing MySQL pessimistic locking (`PESSIMISTIC_WRITE`) and transactional updates.
- **Containerized local services** with Docker Compose (MySQL, Kafka, ZooKeeper) and wrote automated unit tests using JUnit 5 and Mockito to verify system reliability.

EDUCATION

Seattle Pacific University | *Bachelor of Science in Computer Science* Sep 2024 - June 2026

- GPA: 3.89
- Related Coursework: Data Structures & Algorithms, API Design, Systems Design, and Database Management.

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate Mar 2026

JPMorgan Chase & Co. Software Engineering Job Simulation Sep 2026

WORK EXPERIENCE

Math Teaching Assistant | Edmonds College Jan 2023 - June 2024

- Provided 1:1 tutoring to students in the Math Center, helping clarify concepts and solve problems.
- Assisted instructors during classroom sessions and supported students with course material.
- Explained mathematical concepts and problem-solving methods in a clear and approachable way.

TECHNICAL SKILLS

Languages: Java, C++, SQL, HTML, CSS, JavaScript, R

Developer Tools: Git, GitHub, Docker, AWS, Azure, Postman, GitHub Actions, Jira, MySQL Workbench

Libraries & Framework: Spring Boot, Spring Security, Spring Data JPA, React, Bootstrap

Databases & Messaging: MySQL, Apache Kafka

Testing: JUnit, Mockito, Cypress